

Exhibit CC-13

JEDEC Unveils Plans for DDR5 MRDIMM and LPDDR6 CAME Standards to Propel High-Performance Computing and AI

ARLINGTON, Va., USA - July 22, 2024 – JEDEC Solid State Technology Association, the global leader in the development of standards for the microelectronics industry, proudly announces upcoming standards for advanced memory modules designed to power the next generation of high-performance computing and AI applications. JEDEC today revealed key details about its upcoming standards for DDR5 Multiplexed Rank Dual Inline Memory Modules (MRDIMM) and a next-generation Compression-Attached Memory Module (CAME) for LPDDR6. The new MRDIMM and CAME for LPDDR6 are set to revolutionize the industry with unparalleled bandwidth and memory capacity.

DDR5 MRDIMMs offer an innovative, efficient new module design to enhance data transfer rates and overall system performance. Multiplexing allows multiple data signals to be combined and transmitted over a single channel, effectively increasing the bandwidth without the need for additional physical connections and providing a seamless bandwidth upgrade to enable applications to exceed DDR5 RDIMM data rates. Other planned features include:

- Platform compatibility with RDIMM for flexible end-user bandwidth configuration
- Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption
- Efficient I/O scaling using RCD/DB logic process capability
- Leverage existing LRDIMM ecosystem for design and test infrastructure
- Support for Multi-generational scaling to DDR5-EOL

The JEDEC MRDIMM standard is set to deliver up to twice the peak bandwidth of native DRAM, enabling applications to surpass current data rates and achieve new levels of performance. It maintains the same capacity, reliability, availability, serviceability (RAS) features as JEDEC RDIMM. The committee aims to double the bandwidth to 12.8 Gbps and increase the pin speed. MRDIMM is envisioned to support more than two ranks and is being designed to utilize standard DDR5 DIMM components ensuring compatibility with conventional RDIMM systems.

Plans are underway for a Tall MRDIMM form factor to offer higher bandwidth and capacity without changes to the DRAM package. This innovative, taller form factor will enable twice the number of DRAM single-die packages to be mounted on the DIMM without the need for 3DS packaging.

As a follow-on to JEDEC's JESD318 CAME2 Memory Module standard, JC-45 is developing a next-generation CAME module for LPDDR6 targeting a maximum speed greater than 14.4 GT/s. As planned, the module will also offer a 24-bit subchannel, a 48-bit channel and a connector array.

Both projects are in development in JEDEC's JC-45 Committee for DRAM Modules. JEDEC encourages companies to join and help shape the future of JEDEC standards. Membership grants access to pre-publication proposals and provides early insights into active projects such as MRDIMM. [Discover the benefits of membership and join today.](#)

JEDEC standards are subject to change during and after the development process, including disapproval by the JEDEC Board of Directors.

About JEDEC

JEDEC is the global leader in the development of standards for the microelectronics industry. Thousands of volunteers representing over 350 member companies work together in more than 100 JEDEC committees and task groups to meet the needs of every segment of the industry, for manufacturers and consumers alike. The publications and standards generated by JEDEC committees are accepted throughout the world. All JEDEC standards are available for download from the JEDEC website. For more information, visit <https://www.jedec.org>.

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